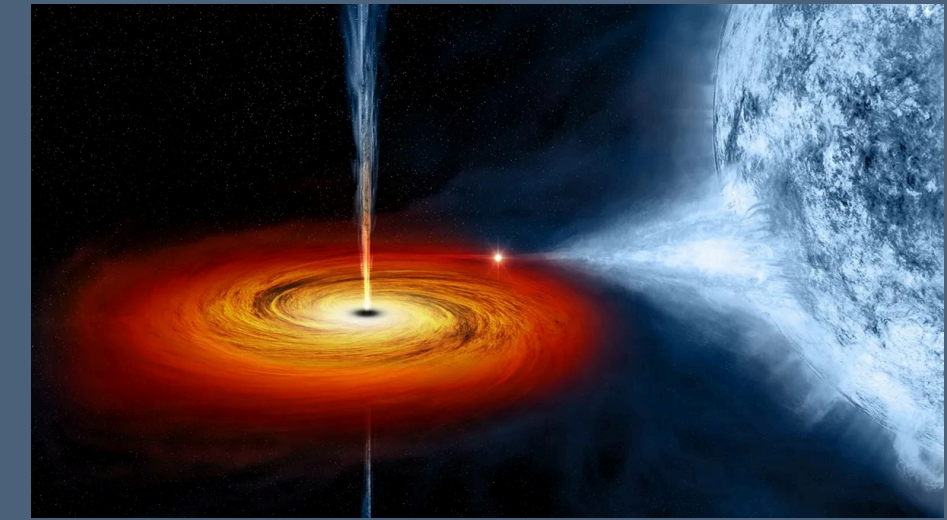


Spurious “dips” near occultation entry

What’s causing them, and what to do about it?

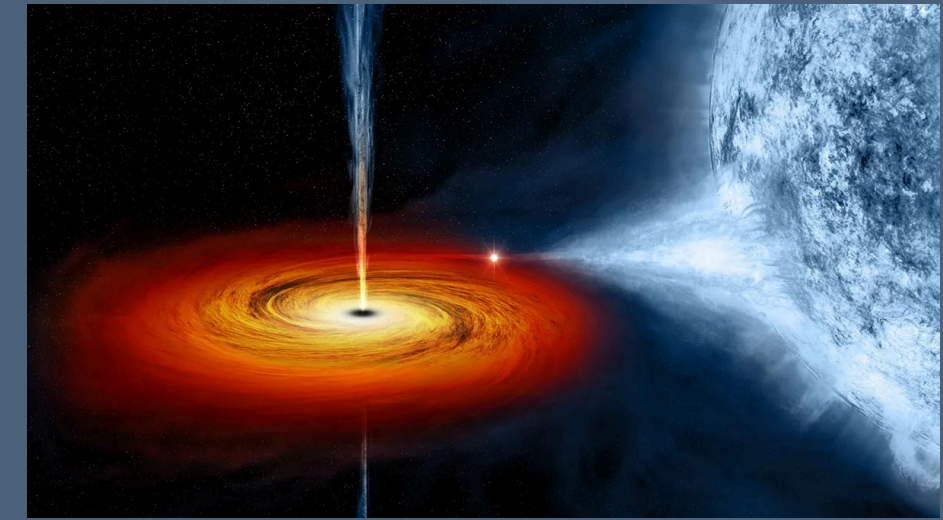
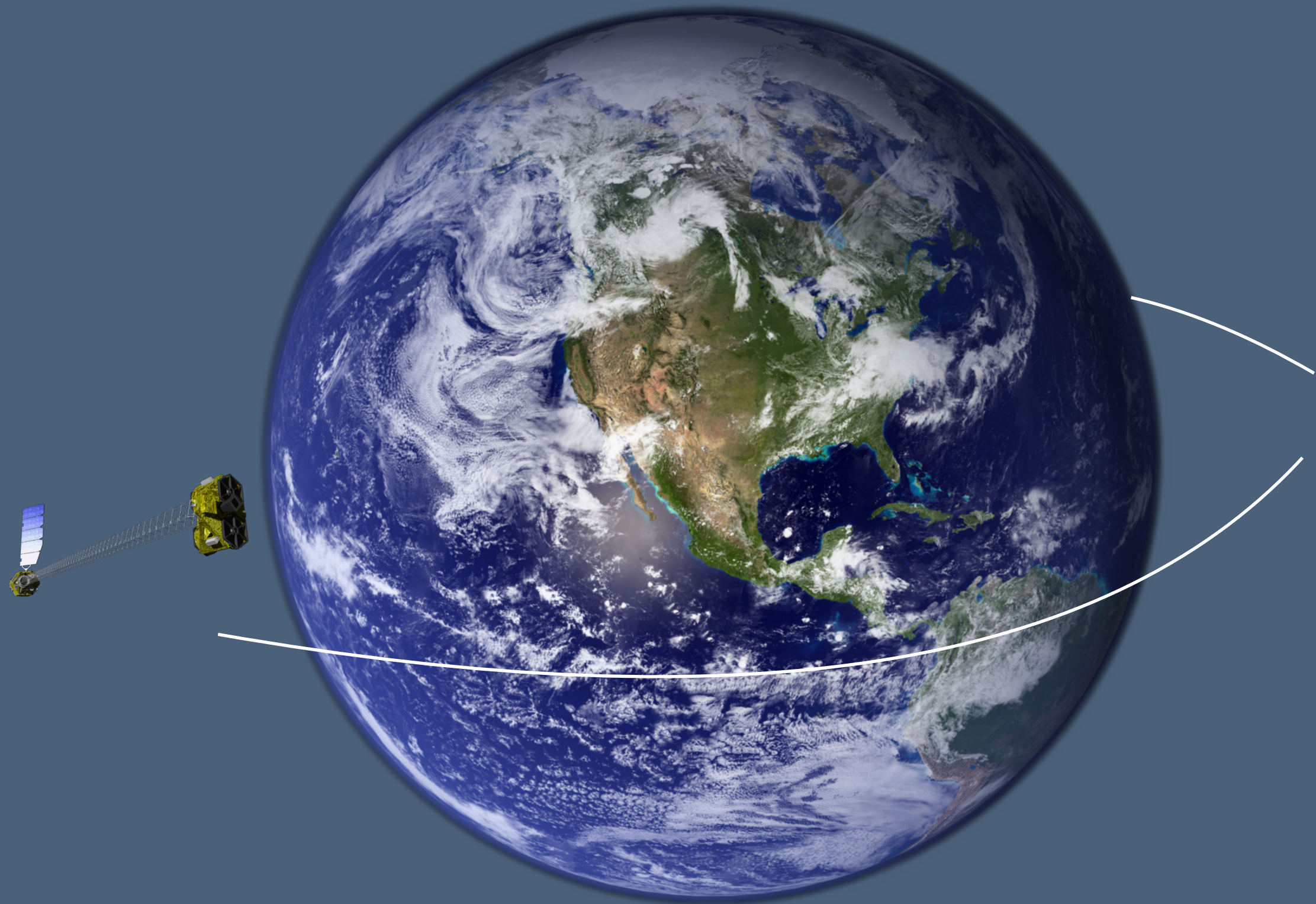
What is occultation?

NuSTAR orbits the Earth once every 96-minutes



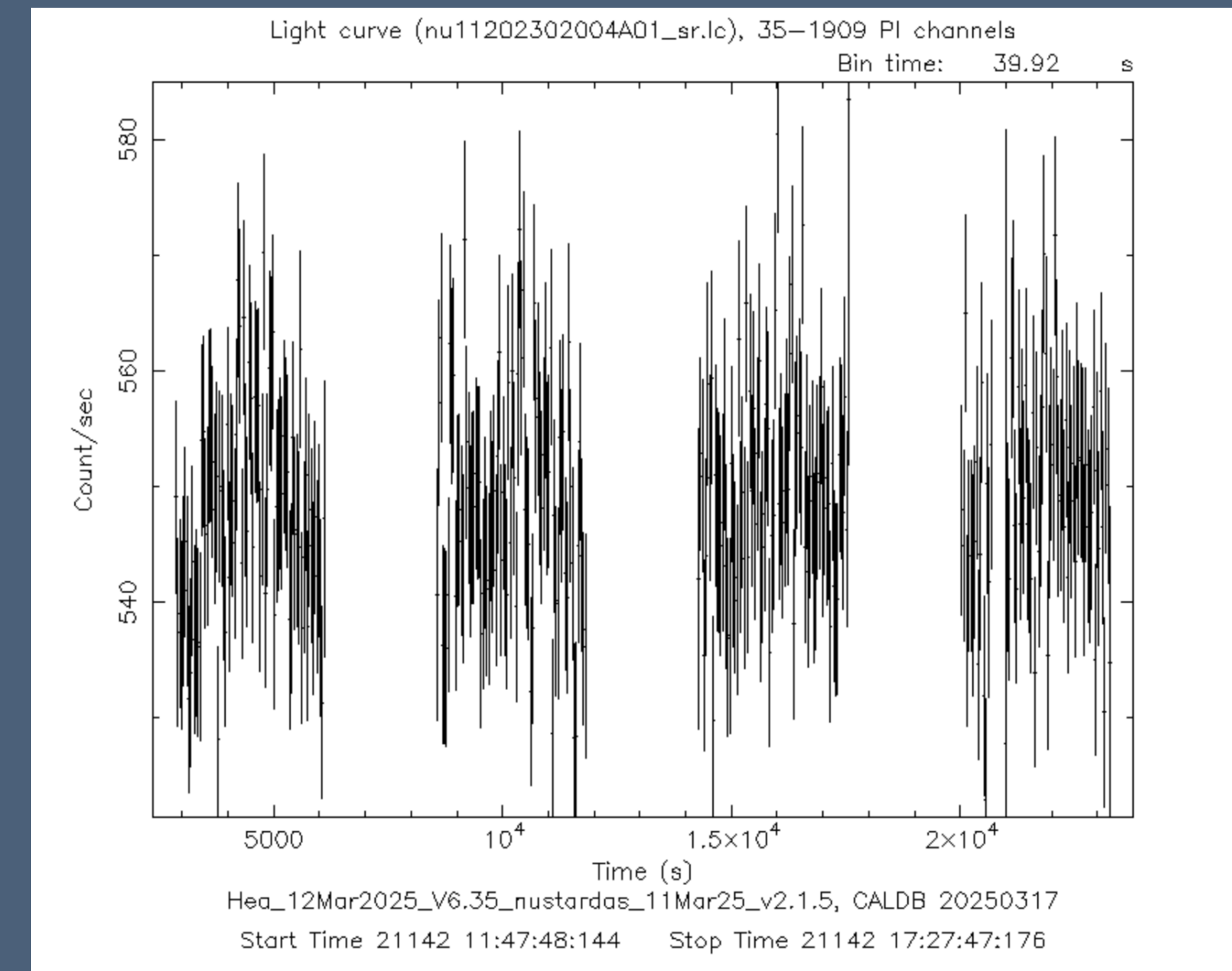
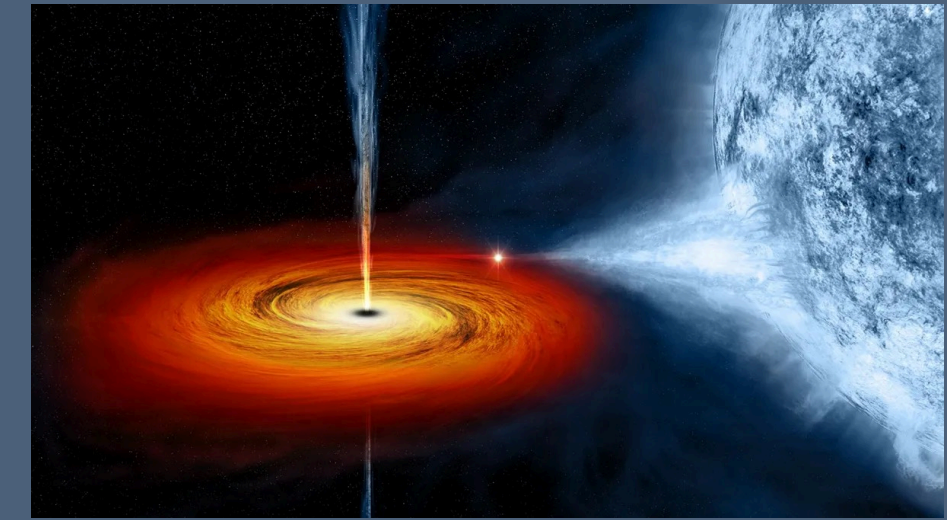
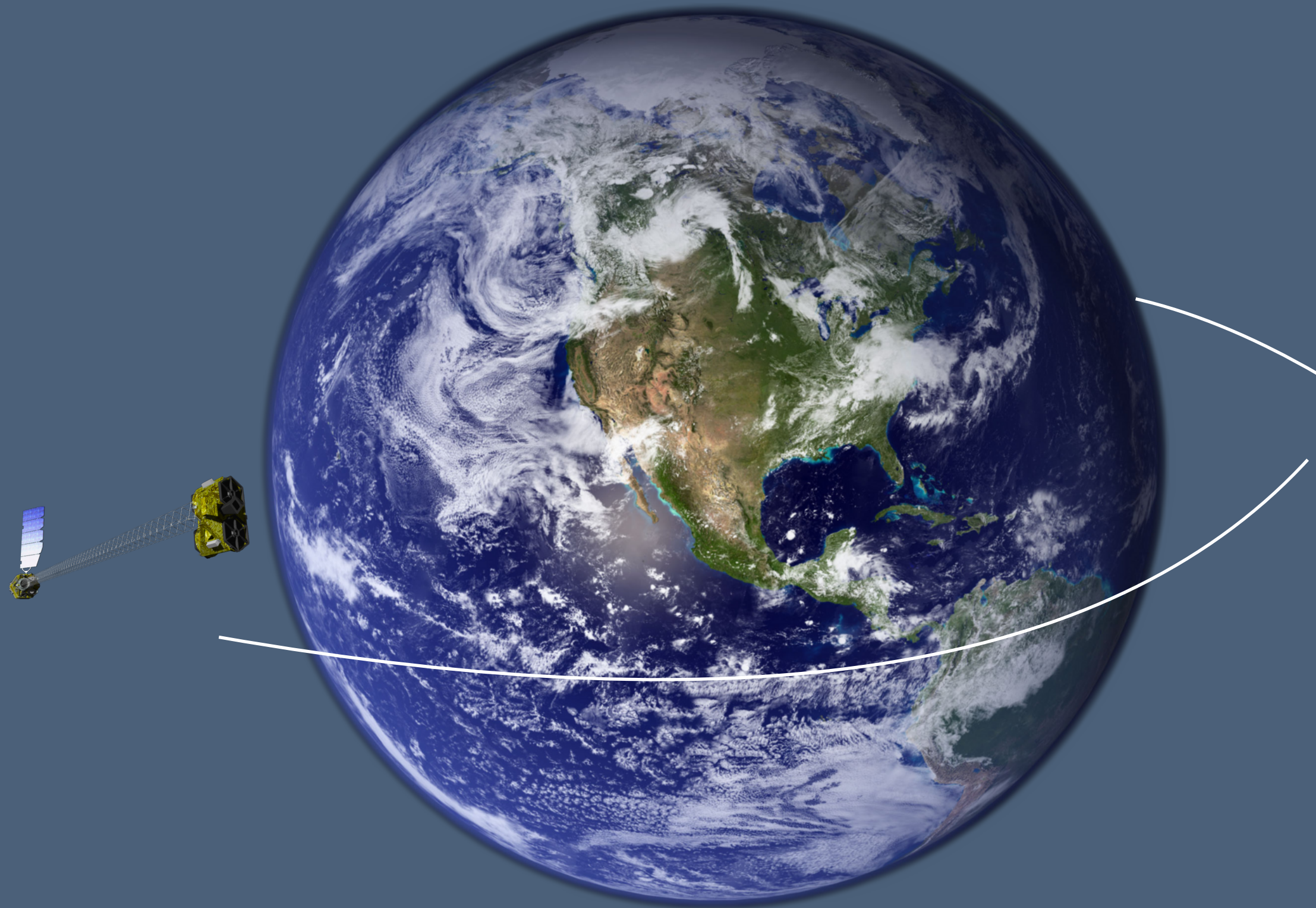
What is occultation?

For part of the orbit, the Earth blocks our view



What is occultation?

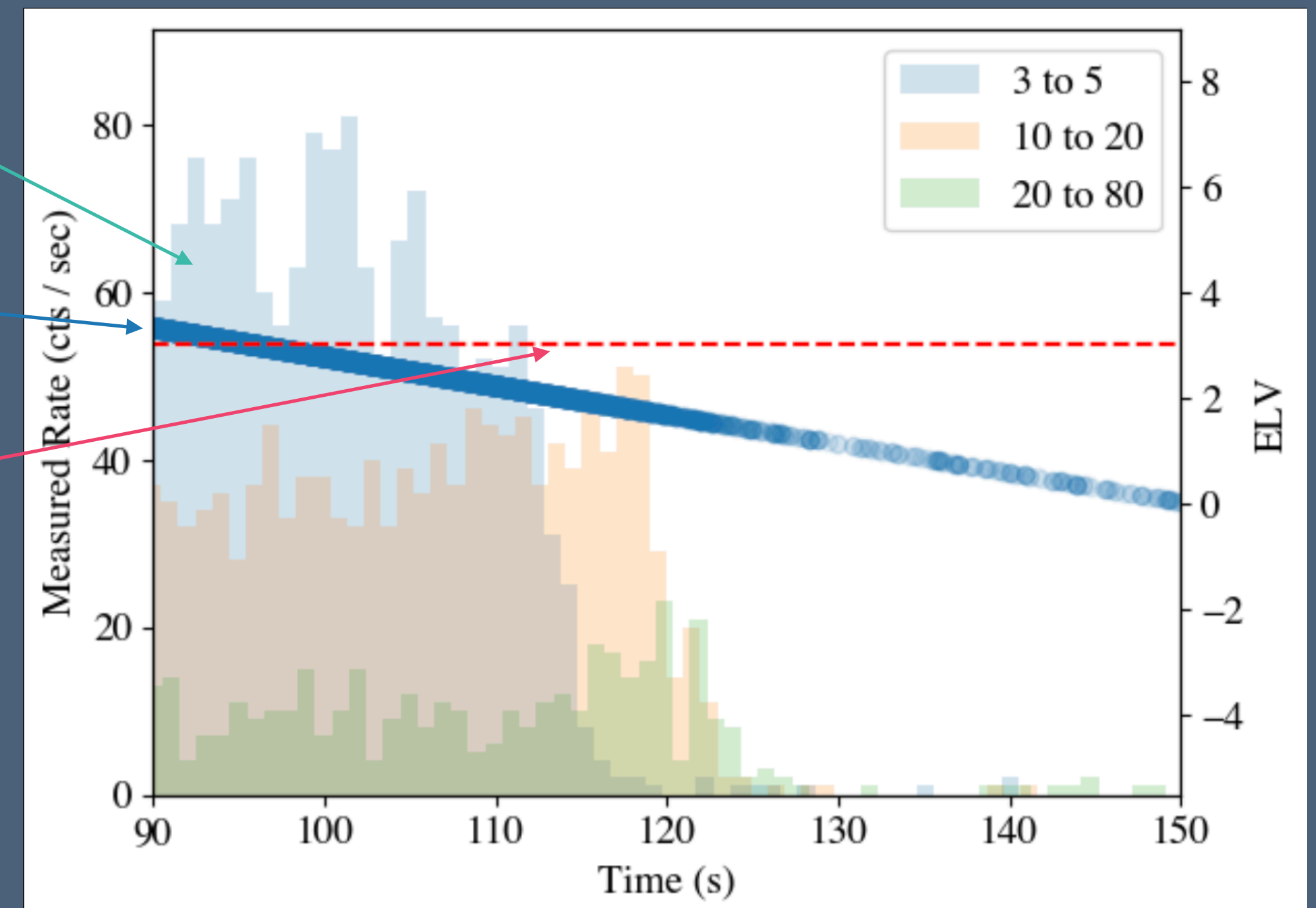
This blocks the X-rays, which result in gaps in the light curve



How do we usually handle this?

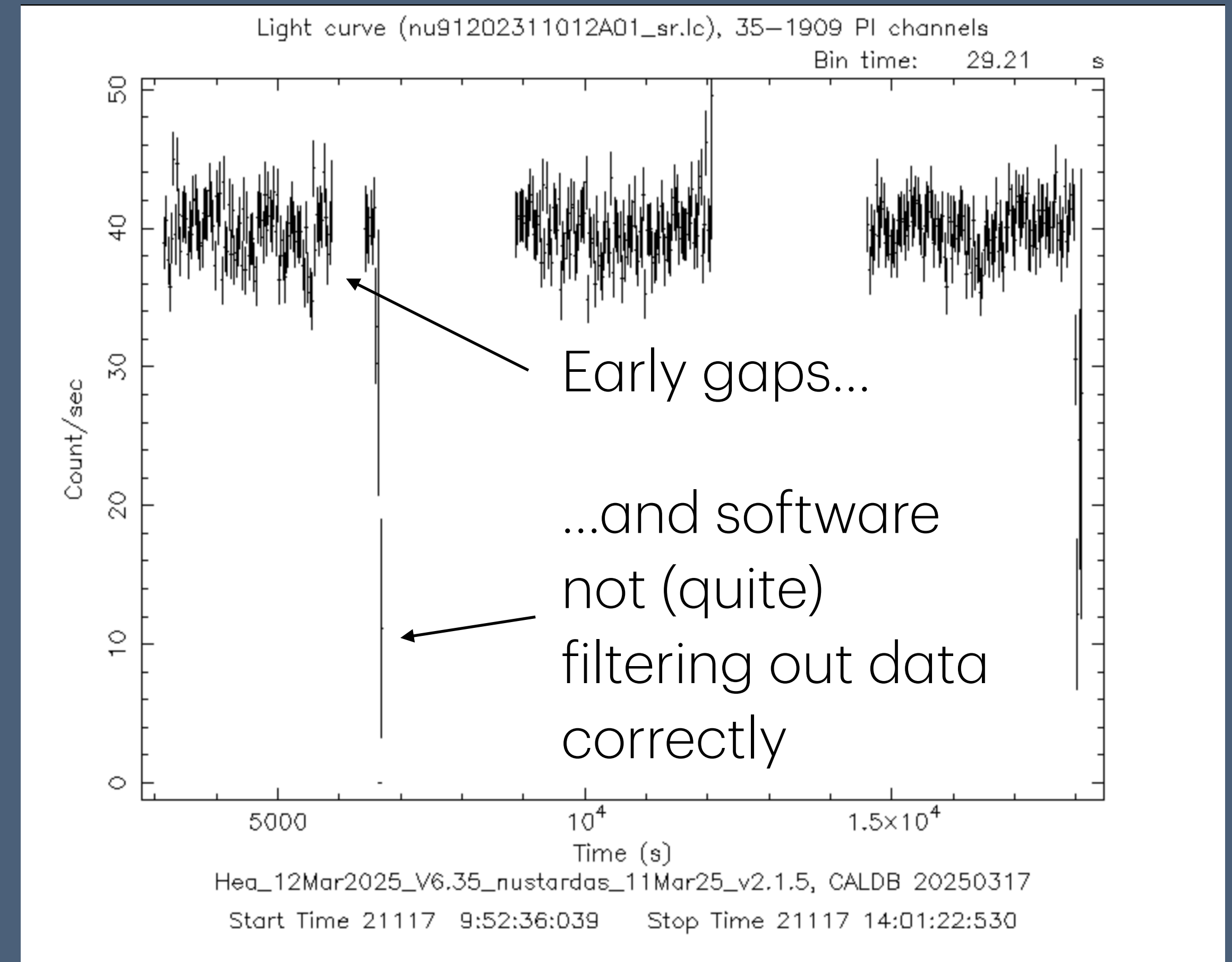
During `nupipeline` runs, `prefilter` FT00L compute ELV to Earth limb

- Data are showing the histogram of EVENTS in various energy bands
- Blue symbols are the ELV vs time
- Horizontal dashed red line shows the **software** cutoff that `nupipeline` uses to split the data into “SCIENCE” data (Mode 01), “OCCULTED” data (Mode 02), and “SCIENCE_SC” (Mode 06) data
- See https://heasarc.gsfc.nasa.gov/docs/nustar/analysis/nustar_swguide.pdf



What's the problem?

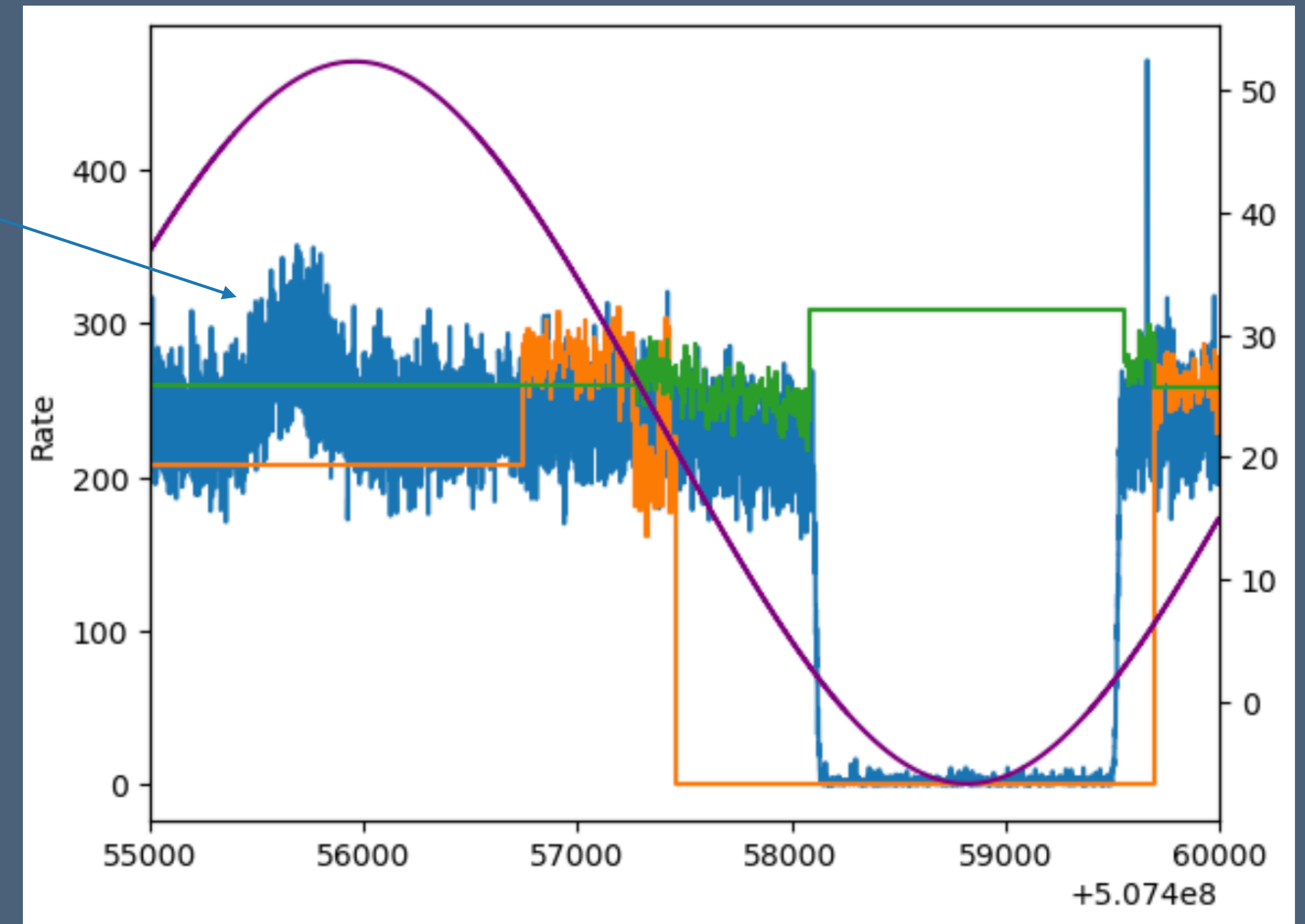
Sometimes things flicker at high declination...



What's happening?

The “CHU4” star tracker is getting blinded by the Earth, but X-rays are still coming through!

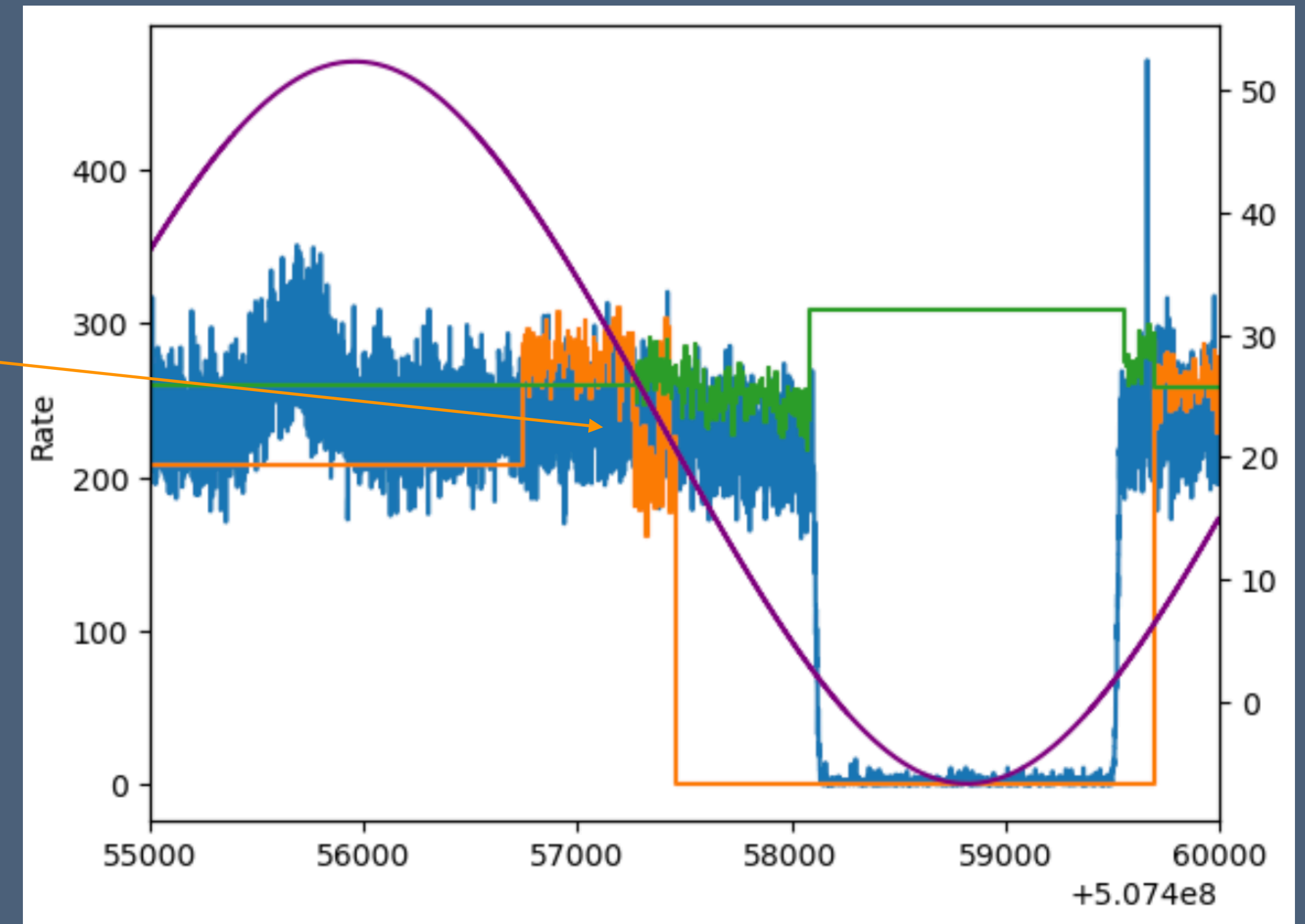
- Complicated plot, it tells the whole story
- Blue is “housekeeping” data rate (or “truth”)
- Orange is “Mode 01” science rate (showing odd dips)
- Green is “Mode 06” science rate (ELV > 3 but no CHU4)
- Purple is “ELV” angle



What's happening?

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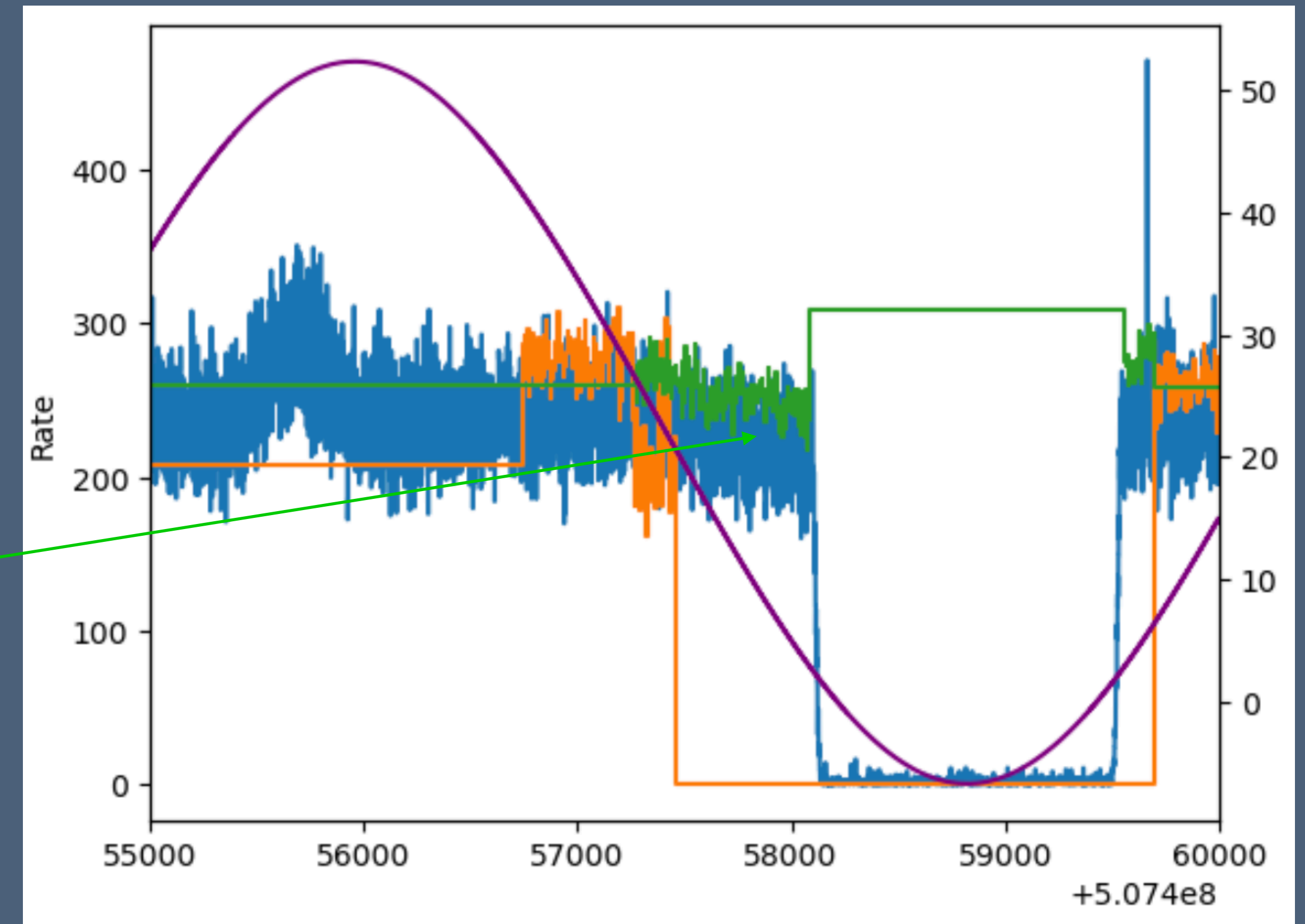
- Complicated plot, it tells the whole story
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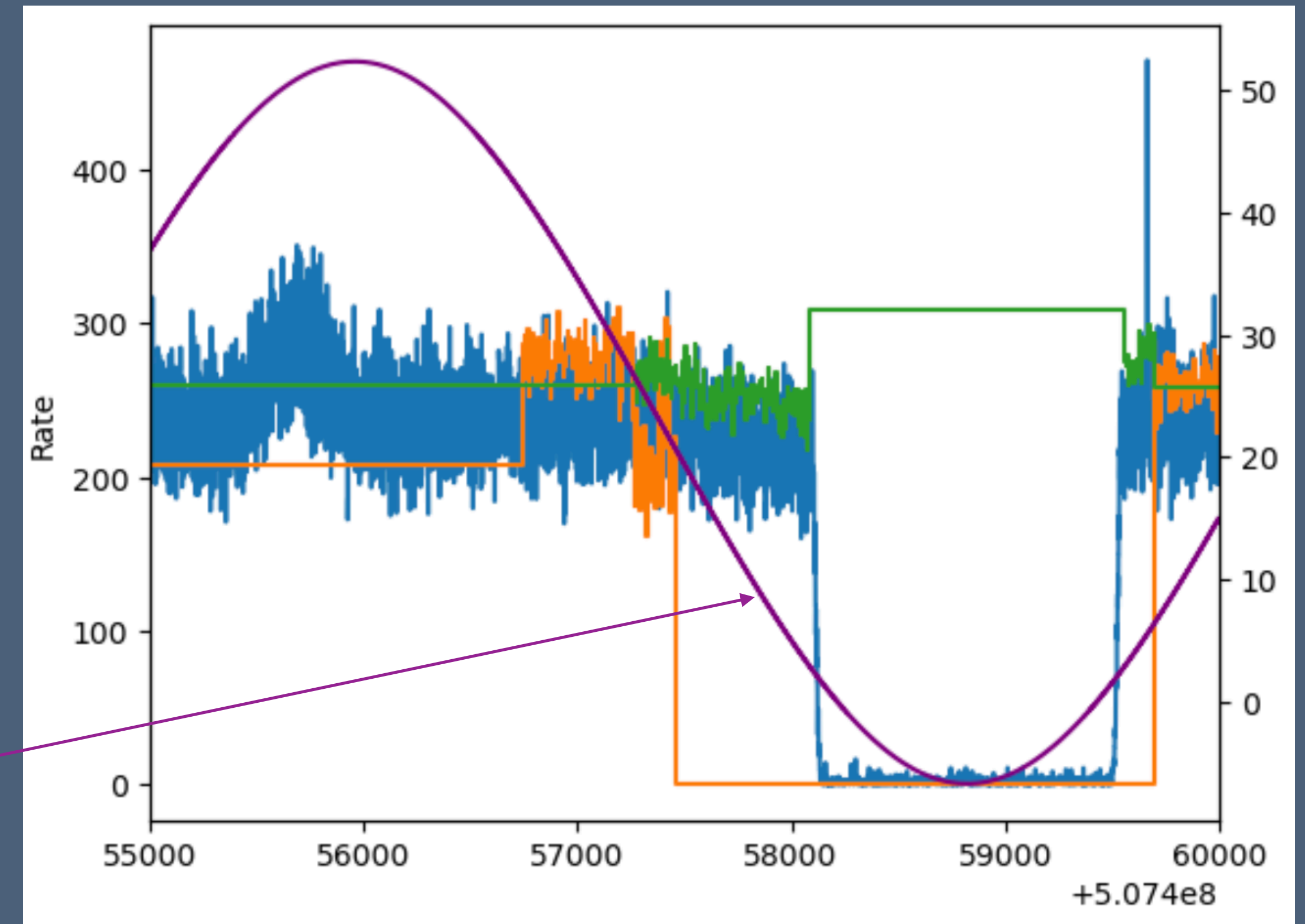
- Complicated plot, it tells the whole story
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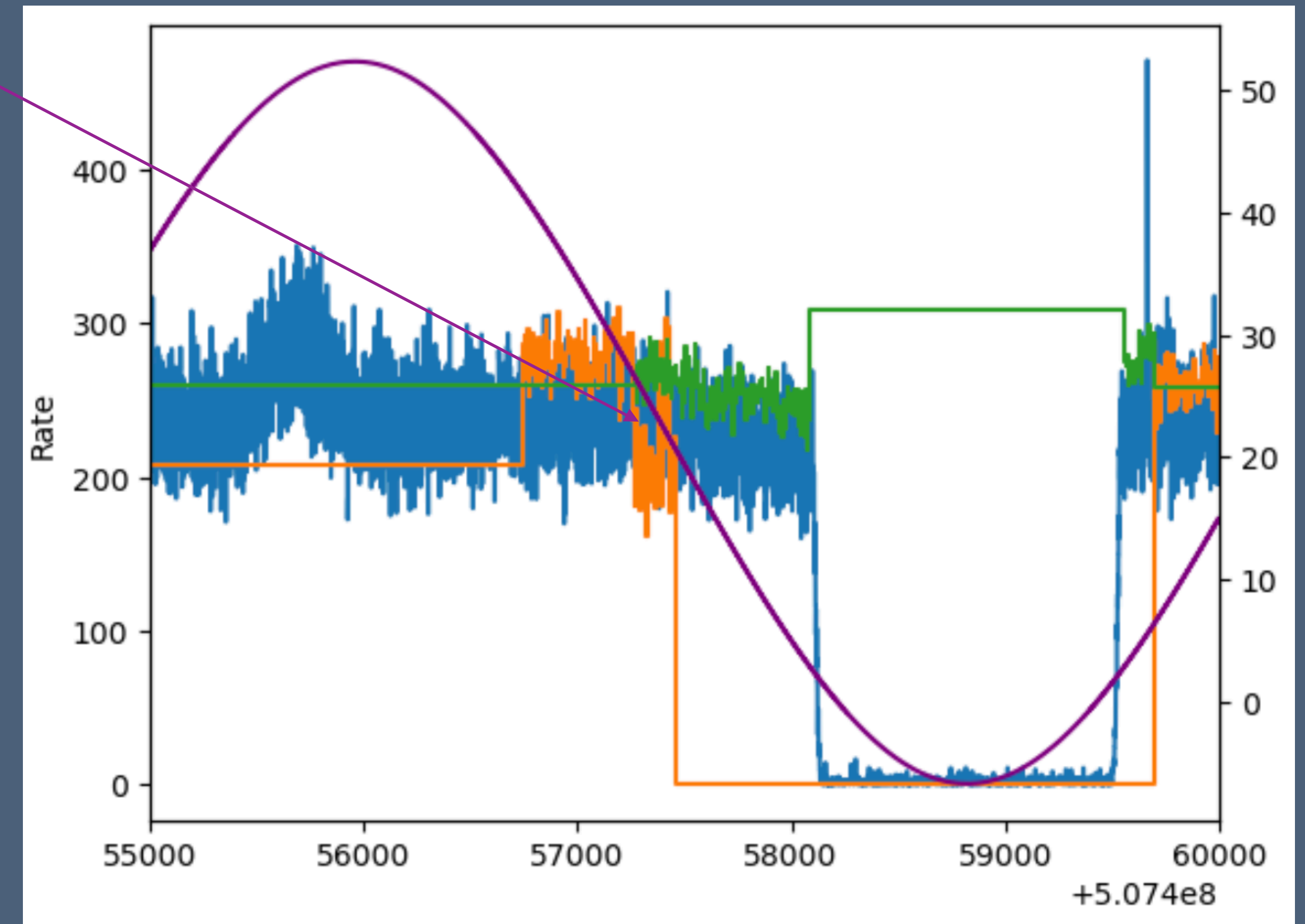
- Complicated plot, it tells the whole story
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What's happening?

The “CHU4” star tracker is getting blinded by the Earth, but X-rays are still coming through!

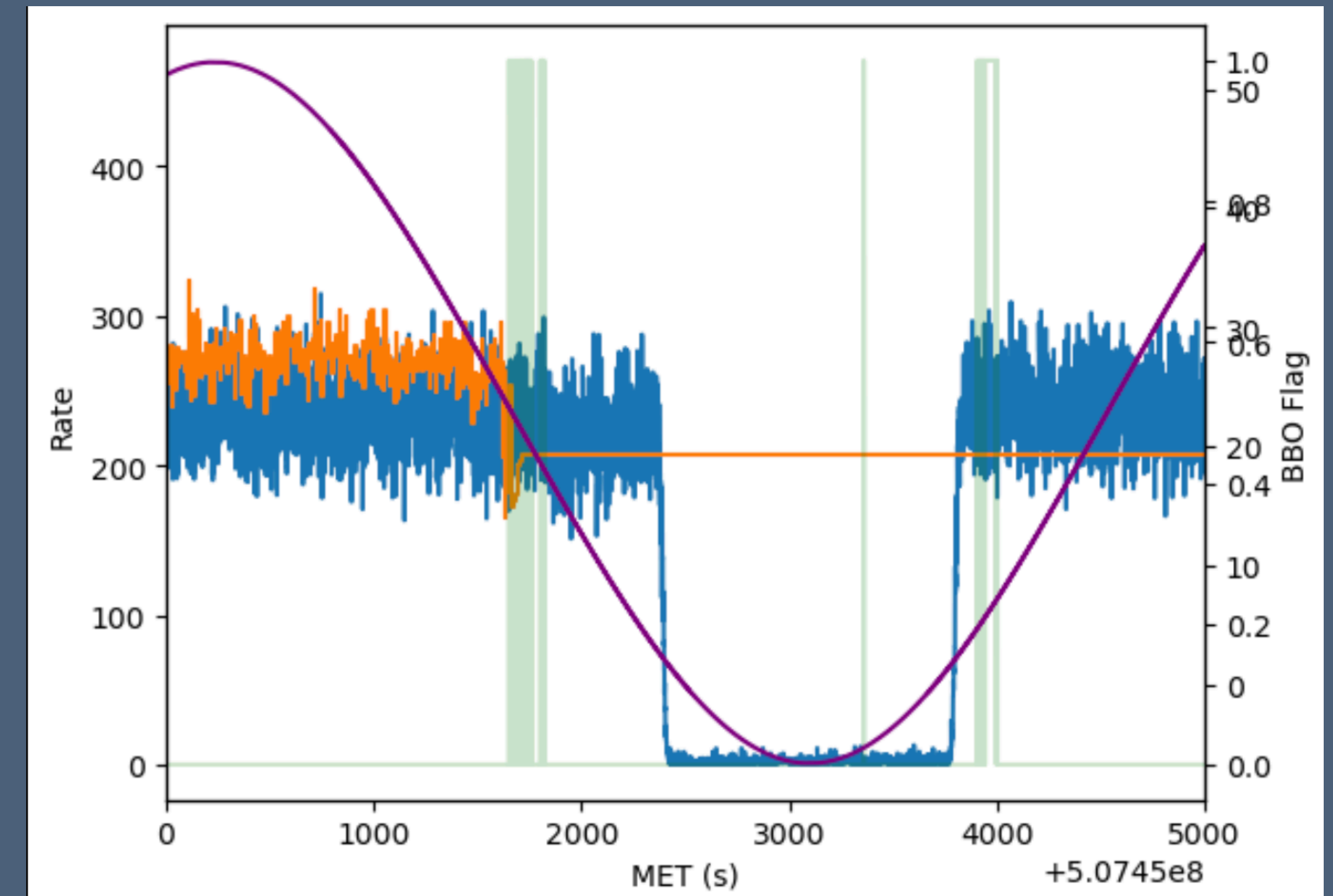
- Very little change in the **actual** data (blue) even though “clean” light curve is behaving badly
- Glint starts around 25-degrees above the Earth limb
- FoV for CHU4 camera is 25-degrees... seeing “glint” off the inside of its baffle (probably)
- Mode 01 + Mode 06 = All Events



What can we do?

Option 1: Filter out flickering periods

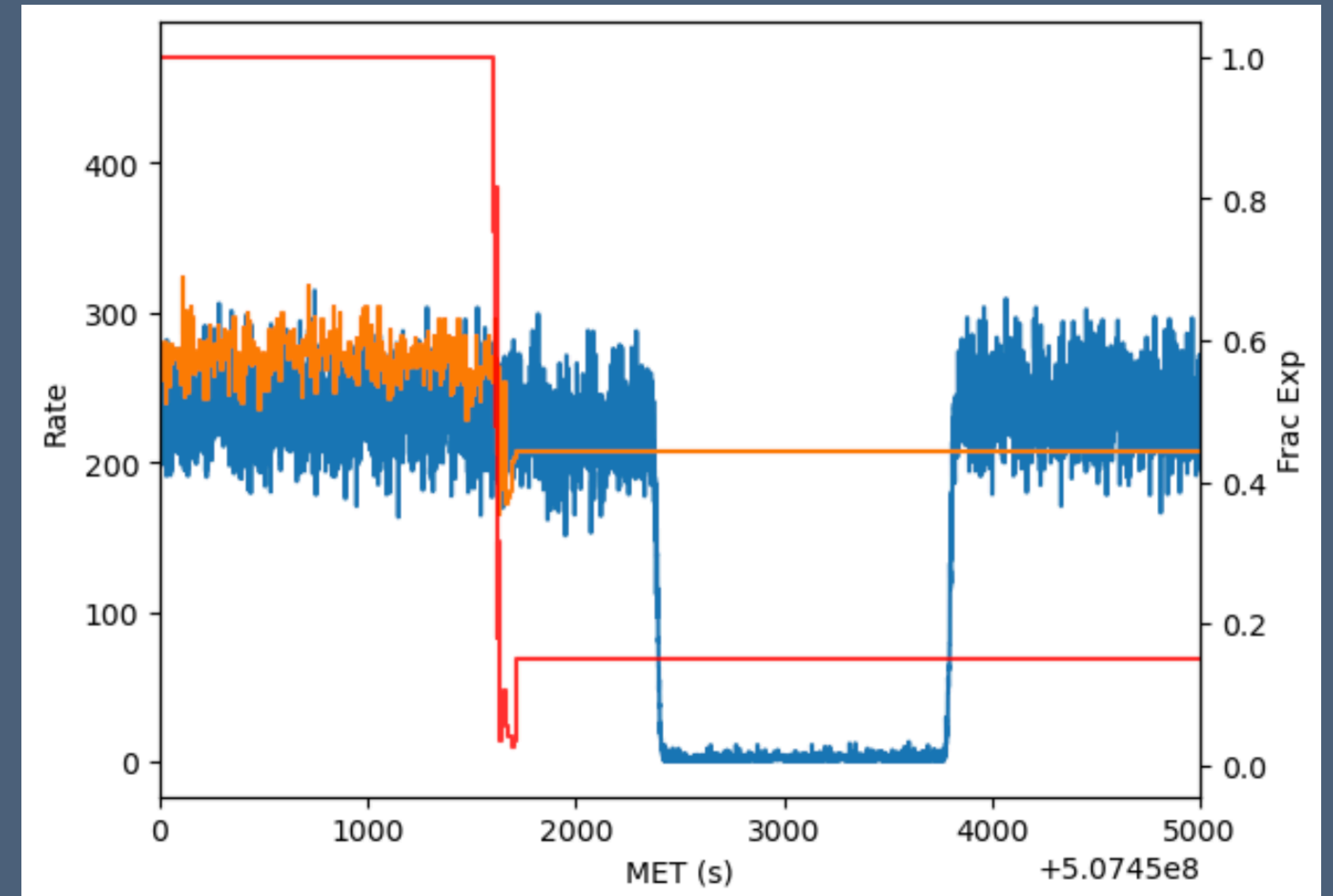
- Green data show “Big Bright Object” (BBO) flag in the data
- This is flickering on and off, resulting in loss of data
- BBO also truncates the Mode 01 data ~500-s before you lose data to the Earth limb



What can we do?

Option 1: Filter out flickering periods

- Red data show `FRACEXP` encoded in the `nu*.lc` file
- Typical argument is to enforce
$$\text{FRACEXP} > 0.9$$
when analyzing light curves

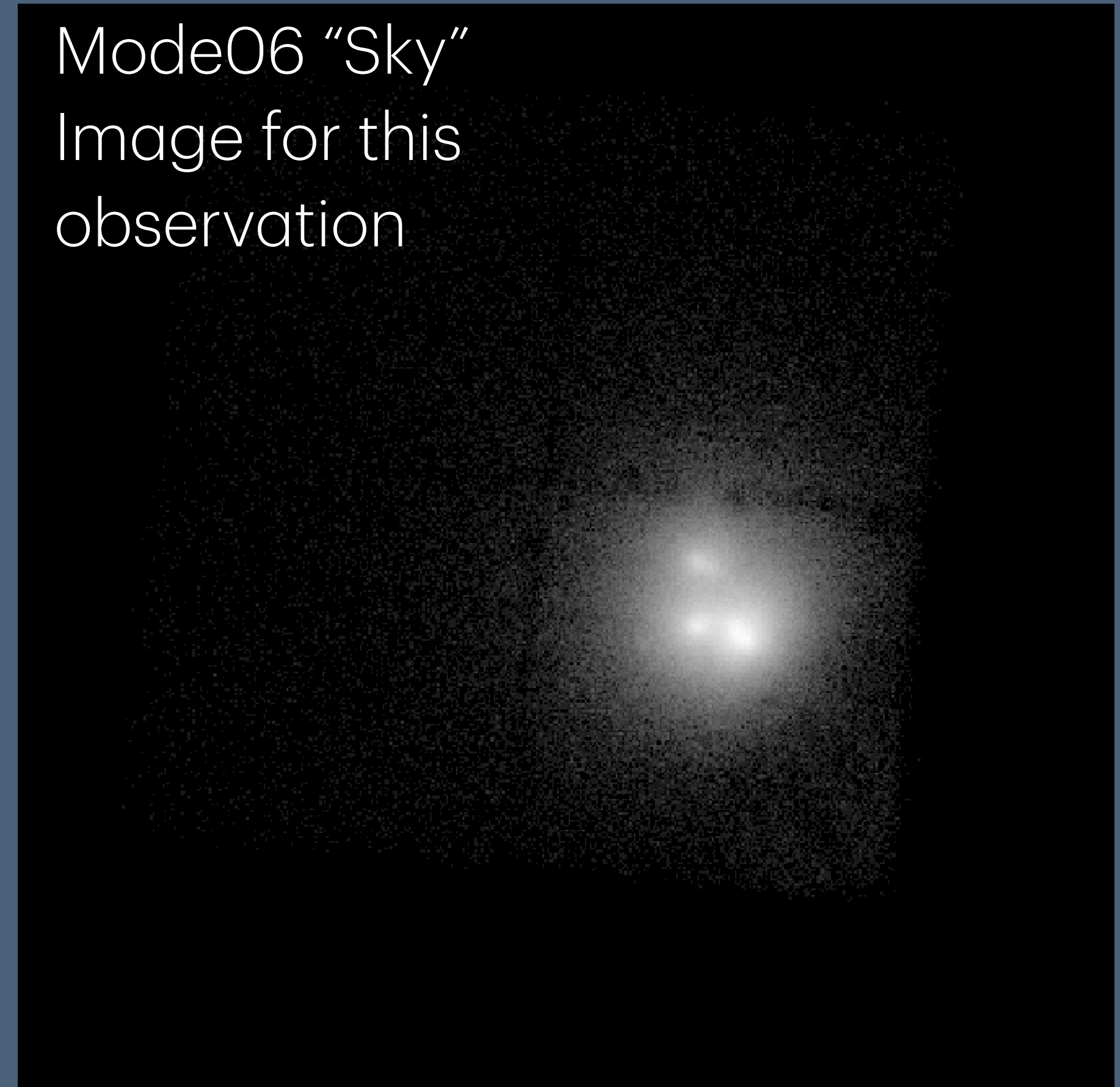


What can we do?

Option 2: Merge 01 and 06 light curves

- The Mode 06 data is a “multi-meatball” because it’s using different CHU combinations
- See appendix of Walton et al (2016) on how to analyze Mode06 data and associated issues
 - <https://iopscience.iop.org/article/10.3847/0004-637X/826/1/87/pdf>
- Some systematics in the light curve products that it’s up to the user to check.
- See `nusplit` in `nupipeline` for how to analyze data split by individual CHU combination (one meatball at a time)

Mode06 “Sky”
Image for this
observation



Summary

- When dealing with **bright sources** at **high declination** can occasionally see some odd glitches in the light curves
- Likely caused by glint in the CHU4 star tracker
- Results in loss of Mode 01 light curve coverage as we go into/out-of occultation
- Probably not noticed for faint sources (when light curves bins are >100-s)
- Can either resolve by properly filtering on FRACEXP value (“standard practice”) or by merging light curves from Mode 01 and Mode 06 (“black belt” analysis, but well understood)